Efficient Firm Dynamics in a Frictional Labor Market

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Motivation: Hiring Has Two Margins

- **Standard Search Models (Mortensen–Pissarides):**
  - Firms hire more workers mainly by posting more vacancies.
  - All vacancies are assumed to be filled at the same aggregate rate $q(\theta)$.

- **Puzzle 1 (Davis, Faberman, and Haltiwanger, 2013):**
  - The job-filling rate almost *doubles* as monthly firm employment growth increases from 10% to 20%.
  - Firms increase hiring through both more vacancies and more effective recruiting.

- **Puzzle 2:**
  - In empirical VARs, job-finding rates and labor market tightness respond slowly to economic shocks (Fujita and Ramey, 2007).
  - But standard models predict that tightness adjusts immediately with productivity (Shimer, 2005).

Three Main Contributions

New Theory

Firms compete for workers by offering long-term contracts.

To grow faster, they offer higher wages; job-filling rates become **firm-specific**.

Efficiency

The model achieves an efficient allocation of workers, unlike bargaining models (Stole-Zwiebel) where large firms may over-hire to push down wages.

Methodological

The model is tractable even outside steady state, allowing the authors to study business-cycle dynamics.

The Microfoundation: How "Recruiting intensity" Works

- **Primitives:**
  - Continuum of risk-neutral workers (mass 1, discount factor β, flow non-work income b).
  - Large heterogeneous firms producing $xzF(L)$ with $F'(L) > 0, F''(L) < 0$.
  - Idiosyncratic productivity x and aggregate productivity z follow Markov processes; entry cost $K(z)$, exogenous exit δ_0, worker quits s_0.
- **The market:**
  - Firms publicly post long-term wage contracts; unemployed workers observe offers and direct search.
  - Within a period: shocks/entry revealed $\rightarrow$ firms post vacancies and contracts $\rightarrow$ matching occurs.
- **Matching Technology & The Price of Speed:**
  - Filling rate m requires applicant queue $\lambda(m) = \tilde{m}^{-1}(m)$ (strictly convex, $\lambda'(0) \geq 1, \lambda'(1) = \infty$).
  - Worker meeting rate $m/\lambda(m)$ is decreasing. Equalizing worker search value yields:

$$w(m) = b + \rho + \frac{1-\beta\varphi}{\beta(1-\delta)} \cdot \frac{\lambda(m)}{m}\rho, \qquad \varphi \equiv (1-\delta)(1-s)$$

where $\rho \equiv (1-\beta)U - b$ is the market-wide flow worker search value pinned down in equilibrium.

Firm's Problem: Two Priced Margins

- **The Firm's Dynamic Value Function:**

$$J_x(L) = \max_{m,\, V \geq 0} \left\{ xF(L) - C(V, L, x) - D(m)V + \beta(1-\delta)\mathbb{E}\left[J_{x'}(L^+)\right] \right\}$$

$$\text{s.t.} \quad L^+ = L(1-s) + mV, \qquad D(m) \equiv w(m)\,\frac{\beta(1-\delta)}{1-\beta\varphi}$$

- **Two Channels, Two Shadow Prices:**
  - **Quantity (V):** Priced by convex recruitment costs (using incumbent staff time):

$$C(V, L, xz) = \frac{c}{1+\gamma}\left(\frac{V}{L}\right)^{\gamma} V, \quad \text{benchmark } \gamma = 2 \text{ (cubic cost)}$$

  - **Speed (m):** Priced by contract wage commitments $D(m)V$ for the mV new hires.
- **Comparative Statics:** Optimal filling rate $m(x, L)$ is strictly increasing in productivity x and decreasing in size L; growing firms offer better contracts (matching Davis et al., 2013).

Phase Diagram

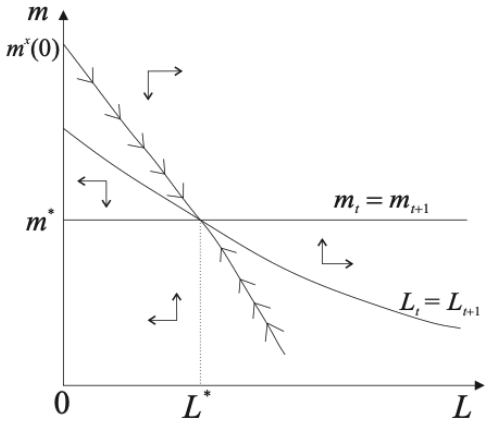


Figure 1: The firm's optimal recruitment policy follows the declining saddle path.

Equilibrium: Four Conditions; Shocks: The Planner's Shortcut

1. The Stationary Competitive Search Equilibrium

An equilibrium is a list $\{\rho, N_0, (L_a^x, m_a^x, V_a^x, w_a^x)\}_{x,a}$ satisfying four conditions:

- **Worker optimality:** The wage-posting mapping $w(m)$ holds across all active submarkets.
- **Firm optimality:** The dynamic recruitment path (V_a^x, m_a^x) solves the firm's problem for each x.
- **Free entry:** $\sum_x \pi(x) J_x(0,0) \leq K$ (equality if $N_0 > 0$); **pins down the search value** ρ.
- **Resource feasibility:** $\sum_{a,x} N_0 (1-\delta)^a \pi(x) \left[L_a^x + \lambda(m_a^x) V_a^x\right] = 1$; **pins down entrant mass** N_0.

2. Efficiency (Modified Hosios): With matching elasticity $\varepsilon_{m,\lambda} \equiv \frac{\lambda(m)}{\lambda'(m)m} \in [0,1]$:

$$C_1(V, L, x) = \frac{1 - \varepsilon_{m,\lambda}}{\varepsilon_{m,\lambda}} \, m \, S_w, \qquad \rho = \frac{m}{\lambda(m)} S_w$$

What Happens with Shocks: Why the Model Stays Solvable

- **Completing Job Flows (Creation & Destruction):**
  - With firm-level and aggregate productivity shocks, firms can also shrink and exit, completing the picture of job creation and destruction.

- **The Core Surprise (Block Recursivity):**
  - Despite full firm heterogeneity, solving the model **does not require tracking the distribution of firms**.
  - A social planner's problem collapses to:
    1. One firm-level Bellman recursion, plus
    2. One low-dimensional fixed point.
  - Free entry simply absorbs the remaining job seekers as a residual.

Cross-Sectional Results: Reproduces Davis et al. Patterns:

- Bigger growth goes with more vacancies and higher filling rates, and fast-growing firms pay higher wages to comparable workers

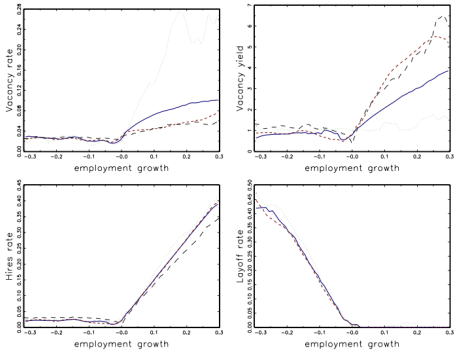


Figure 3: Cross-sectional relationships between monthly employment growth and the vacancy rate, the vacancy yield, the hires rate and the layoff rate. The dashed curves (in the first three graphs) are from the data used in Davis et al. (2013), the blue (solid) curves are for the model with cubic recruitment costs ($\gamma = 2$), the green (dotted) curves are for $\gamma = 0.1$ and the red (closely dashed) curves are for $\gamma = 8$. Model statistics are calculated from a cross-section of $4.9 \cdot 10^6$ firms.

Business Cycle Results: Slow Adjustment

- **Convex recruiting costs** generate procyclical recruiting intensity and delayed, hump-shaped responses in job-finding and market tightness, resolving the jump-variable problem.

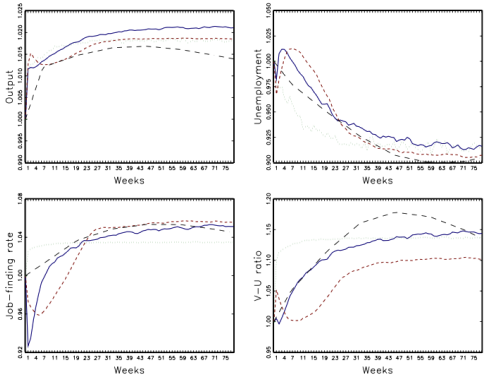


Figure 4: Impulse response to a permanent 1% increase in aggregate productivity. The dashed curves are responses from a VAR for U.S. data (see Appendix E for details), the blue (solid) curves are for the model with cubic recruitment costs ($\gamma = 2$), the green (dotted) curves are for $\gamma = 0.1$ and the red (closely dashed) curves are for $\gamma = 8$. All three versions have unemployment income $b \approx 0.977 \cdot w$.

Policy Punchline and Conclusion

- **Policy Experiment: Hiring Subsidies (e.g., HIRE Act 2010):**
  - they **backfire**
  - Free entry pushes the subsidy into higher wages.
  - Higher wage commitments make labor hoarding costly $\rightarrow$ firms **fire more workers** during downturns.
  - Separation-rate volatility rises from 8.5% to 11.2%; steady-state unemployment increases.

- **Conclusion**
  1. Wages function as a direct recruiting tool to purchase hiring speed.
  2. Competitive search preserves constrained efficiency with multi-worker firms.
  3. Policy that ignores the recruiting-intensity margin can move the wrong margin.